

Capstone Project

Project Title

E-Commerce Automation Testing Using Playwright (Advantage Online Shopping)

Project Overview

This project focuses on automating testing of an e-commerce website using Playwright. It validates key user workflows such as authentication, product search, cart operations, checkout process, wishlist, and user profile management to ensure the application is stable and reliable.

Website

Advantage Online Shopping

URL: <https://www.advantageonlineshopping.com/>

Services Covered

Authentication Service

- User registration with valid and invalid data
- Login with valid and invalid credentials
- Logout functionality
- Forgot password flow
- Session validation

Search Service

- Search with valid product name
- Search with partial keywords
- Search with invalid product
- Auto-suggestions validation
- Empty search handling

Product Service

- Product listing validation
- Product details verification
- Image display validation
- Price accuracy check
- Product variation selection

Cart Service

- Add product to cart
- Remove product from cart
- Update quantity
- Cart total validation
- Cart persistence after refresh

Checkout Service

- Proceed to checkout
- Address selection/add new address
- Payment page navigation
- Order summary validation
- Place order flow

User Profile Service

- View profile details
- Update profile information
- Password change
- Address management
- Profile persistence

Wishlist Service

- Add/remove wishlist items
- Move wishlist item to cart
- Wishlist persistence after login

Order Management Service

- View order history

- Order details validation
- Order status check
- Reorder functionality

API / Mocking Service

- Product API validation
- Cart API validation
- Order API validation
- Status code checks
- Data consistency UI vs API

Tools & Technologies

Category	Technology
Automation Tool	Playwright
Language	JavaScript
Runtime Environment	Node.js
IDE	VS Code
Version Control	Git & GitHub

RTM (Requirement Traceability Matrix)

Requirement ID	Module	Test Scenario	Test Case Count
REQ-1	Authentication	User Login and Registration	18
REQ-2	Search	Search Product Validation	12

REQ-3	Product	Product Details Verification	14
REQ-4	Cart	Cart Operations	18
REQ-5	Checkout	Checkout Flow and Payment	20
REQ-6	Profile	Profile Management	12
REQ-7	Wishlist	Wishlist Validation	10
REQ-8	Order	Order History and Tracking	12
REQ-9	API	API Validation	14

Total Test Cases

120+ Test Cases

Conclusion

This project provides end-to-end automation testing of an e-commerce application using Playwright. It ensures better test coverage, reliability, and supports real-world QA automation practices.